

Synergistic Tuning of Doping Efficiency and Charge Transport in n-Type Thermoelectric Polymers via Rational Backbone Design

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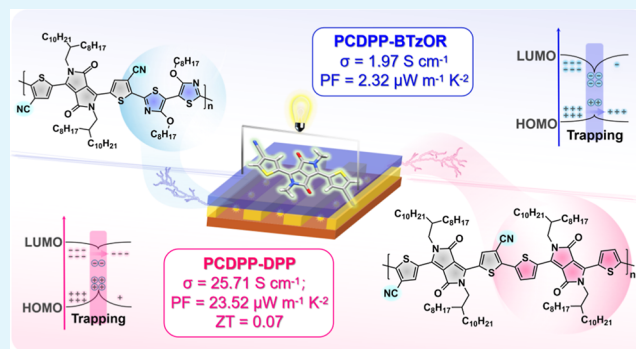
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ABSTRACT: Despite the growing interest in organic thermoelectrics, n-type polymers generally exhibit lower performance than their p-type counterparts, limiting the development of efficient thermoelectric generators. Herein, we report two new n-type polymers based on cyanothiophene-flanked diketopyrrolopyrrole (CDPP) copolymerized with either diketopyrrolopyrrole (PCDPP-DPP) or dialkoxybithiazole (PCDPP-BTzOR) to investigate the critical yet scarcely explored role of orbital interaction between counits in governing thermoelectric performance. Although PCDPP-BTzOR exhibits a more conformationally locked backbone, higher molecular weight, comparable film crystallinity, and nearly identical LUMO level to PCDPP-DPP, the latter displays markedly superior charge transport and doping response. PCDPP-DPP achieves higher field-effect electron mobility (0.191 vs 0.013 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$) and carrier concentration after n-doping (1.4 vs 1.0×10^{20} spins cm^{-3}), yielding over an order of magnitude higher electrical conductivity (25.71 vs 1.97 S cm^{-1}). These improvements arise from the deliberate tuning of interunit orbital interactions in PCDPP-DPP, which promotes electronic delocalization and mitigates trapping associated with polarized charge-transfer states. Consequently, PCDPP-DPP attains a high power factor of 23.52 $\mu\text{W m}^{-1} \text{K}^{-2}$ and a promising room-temperature figure of merit of $ZT = 0.07$. Overall, this work establishes orbital interaction engineering as an effective strategy for designing high-performance n-type thermoelectric polymers.

KEYWORDS: n-type polymer semiconductors, cyano-functionalization, thiophene diketopyrrolopyrrole, organic thin-film transistors, organic thermoelectrics



1. INTRODUCTION

Over the past decades, the growing environmental and energy crisis has intensified global efforts for developing clean and sustainable energy technologies.^{1,2} In this context, π -conjugated polymers (CPs) have emerged at the forefront of next-generation energy conversion technologies due to their potential for manufacturing low-cost, nontoxic, flexible, and stretchable electronic devices via solution-based processing techniques.^{3,4} Beyond their well-established roles in optoelectronics, CPs intrinsically possess low thermal conductivity ($\kappa < 1$ $\text{W m}^{-1} \text{K}^{-1}$) owing to their semicrystalline nature,⁵ making them particularly attractive for organic thermoelectric (OTE) applications where thermal energy from natural or waste heat sources can be directly converted into electricity via the Seebeck effect.^{6–8} This intrinsic characteristic favors high energy conversion efficiencies, quantified by a dimensionless thermoelectric figure-of-merit, $ZT = S^2\sigma T/\kappa$, where S is the Seebeck coefficient, σ is the electrical conductivity, and T is the absolute temperature. Given that κ is already minimized in CPs, the central challenge in this field lies in maximizing the numerator, known as the power factor ($\text{PF} = \sigma S^2$).⁹ However,

the adverse interdependence between σ and S brings significant challenges for optimizing thermoelectric performance in OTEs,¹⁰ as it requires the simultaneous optimization of charge carrier mobility (μ) and carrier concentration (n) upon doping (both governing σ according to $\sigma = \mu ne$, where e is the elementary charge).^{11,12}

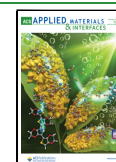
While innovations in materials design and doping strategies have driven remarkable breakthroughs in p-type (hole-transporting) polymers such as PBTTT,¹³ P3HT,¹⁴ and PEDOT:PSS,¹⁵ which routinely exhibit σ values exceeding 1000 S cm^{-1} , only a few n-type (electron-transporting) polymers have achieved σ s over 100 S cm^{-1} due to the persistent challenges of attaining high electron mobilities (μ_e) and the difficulty in achieving efficient n-doping.^{9,16} This

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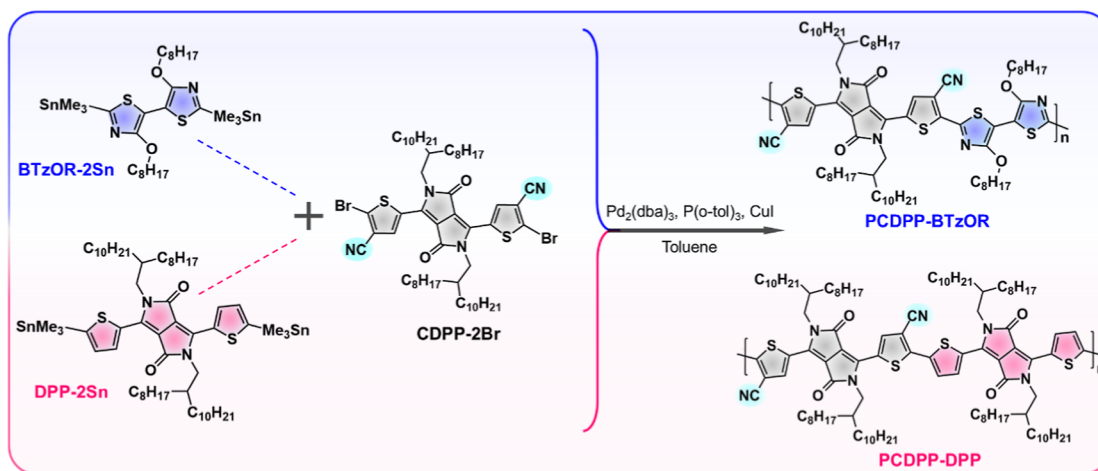
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Scheme 1. Synthetic Route to Polymers PCDPP-BTzOR and PCDPP-DPP Studied in This Work



bottleneck arises from two primary factors: (i) the scarcity of highly electron-deficient building blocks capable of effectively stabilizing the frontier molecular orbitals (FMOs), including both the lowest unoccupied molecular orbital (LUMO) and the highest occupied molecular orbital (HOMO), to facilitate efficient n-doping and electron injection/generation; (ii) the challenge of designing molecular structures that combine backbone rigidity and planarity to promote charge delocalization without compromising solubility.¹⁷ However, even when these conventional design criteria are met, n-type polymers often underperform, suggesting that additional electronic factors may play a decisive role in governing charge transport and thermoelectric efficiency. As a consequence, the intrinsically n-doped poly(benzodifurandione) (PBFDO; LUMO level = −5.18 eV) remains the only n-type CP exhibiting σ values comparable to the state-of-the-art p-type ones, but the limited control over its intrinsic n-doping continues to constrain its thermoelectric performance ($ZT \approx 0.25$).¹⁸ This imbalance between n-type and p-type polymers dramatically hinders not only the development of efficient thermoelectric generators but also for other energy conversion technologies that rely on p–n heterojunctions, which fundamentally require high and well-matched σ s for both hole and electron carriers.^{19–21} Meeting these requirements calls for the design and synthesis of novel n-type CPs with optimized physicochemical properties that enable bridging this gap and the elucidation of structure-property correlations to advance the field.²² One promising strategy involves the chemical modification of weakly electron-accepting units to enhance their electron-withdrawing capabilities without sacrificing structural simplicity or processability.^{23–27} Among these, thiophene-flanked diketopyrrolopyrrole (TDPP) derivatives have attracted significant attention due to their planar π -conjugated framework, favorable π – π interactions, excellent solubility conferred by *N*-alkylation, and facile functionalization.^{28,29} However, the presence of two electron-donating thiophene moieties within the TDPP unit, combined with the delocalization of the amide nitrogen lone pairs along the polymer backbone, leads to relatively high-lying FMO energy levels of the resulting TDPP-based CPs, typically yielding ambipolar or even preferentially p-type transport characteristics.³⁰ This dilemma has recently drawn significant attention, motivating the design of a variety of electron-deficient TDPP derivatives via strategic incorporation of strongly electron-

withdrawing heteroarenes, including pyridine-substituted DPP,³¹ thiazole-flanked DPP,³² and pyrazine-flanked DPP,³³ as well as direct cyano functionalization of the flanked thiophenes.³⁴ Despite these efforts, polymers derived from these building blocks still fall short in achieving high σ values. For instance, although the pyrazine-flanked DPP polymer PzDPP-CT2 and the cyano-functionalized TDPP polymer PCTDPP-BTI both exhibit highly coplanar backbones and remarkably low-lying LUMO/HOMO levels (−3.85/−5.70 eV and −3.90/−5.87 eV, respectively), both display only modest n-doped σ s (8.4 and 4.1 S cm^{−1}) in OTEs.^{33,34} These results highlight the delicate trade-off between enhancing electron affinity of DPP-based units and preserving the optimal structural and electronic characteristics of the resulting polymers, a vaguely explored but ongoing challenge requiring deeper insights into structure-property relationships. In fact, this knowledge gap mirrors a broader trend in the field, where research efforts have largely focused on maximizing molecular planarity and lowering FMO energy levels of comonomers, while often overlooking the equally critical need to ensure electronic compatibility between comonomer units for efficient electron transport along the polymer backbone. Therefore, advancing our understanding of these interconnected factors is essential for the rational design of next-generation n-type CPs with both favorable electronic structures and superior charge-transport properties.

Driven by this remaining challenge, we rationally designed and synthesized two new n-type polymers, PCDPP-BTzOR and PCDPP-DPP, through the copolymerization of cyanothiophene-flanked diketopyrrolopyrrole (CDPP) with either dialkoxy bithiazole (BTzOR) or diketopyrrolopyrrole (DPP) counits. Among them, PCDPP-BTzOR displays a more conformationally locked backbone, higher molecular weight, comparable film crystallinity, and nearly identical LUMO level to PCDPP-DPP; nevertheless, the latter intriguingly shows markedly superior n-type charge transport and thermoelectric properties. The key to this performance lies in the strategic backbone design that enables effective orbital hybridization between counits, thereby promoting electron delocalization while mitigating trapping associated with polarized charge-transfer states. Consequently, PCDPP-DPP delivers a high μ_e of 0.191 cm² V^{−1} s^{−1} in organic thin-film transistors (OTFTs), far exceeding that of PCDPP-BTzOR (0.013 cm² V^{−1} s^{−1}), as well as significantly improved thermoelectric properties ($\sigma = 25.71$

Table 1. Summary of the Molecular Weights, Optical and Electrochemical Properties of Polymers PCDPP-BTzOR and PCDPP-DPP

polymer	M_n [kDa]	D	λ^{sol} [nm] ^a	λ^{film} [nm] ^b	λ_{onset}^{film} [nm]	E_g^{opt} [eV] ^c	E_{LUMO} [eV] ^d	E_{HOMO} [eV] ^d
PCDPP-BTzOR	47.6	2.7	956	944	1106	1.12	−3.75	−5.60
PCDPP-DPP	39.3	2.1	938	932	1086	1.14	−3.76	−5.76

^aAbsorption in dilute chloroform solutions (10^{-5} M). ^bAbsorption of as-cast films. ^cOptical band gap derived from the absorption onset of polymer films using the equation: $E_g^{opt} = 1240/\lambda_{onset}^{film}$ (eV). ^dDerived from CV measurements using the equations: $E_{LUMO} = -(E_{red}^{onset} + 4.8)$ eV, $E_{HOMO} = -(E_{ox}^{onset} + 4.8)$ eV. The reduction (E_{red}^{onset}) and oxidation (E_{ox}^{onset}) potentials were electrochemically determined versus the Fc/Fc⁺ redox couple.

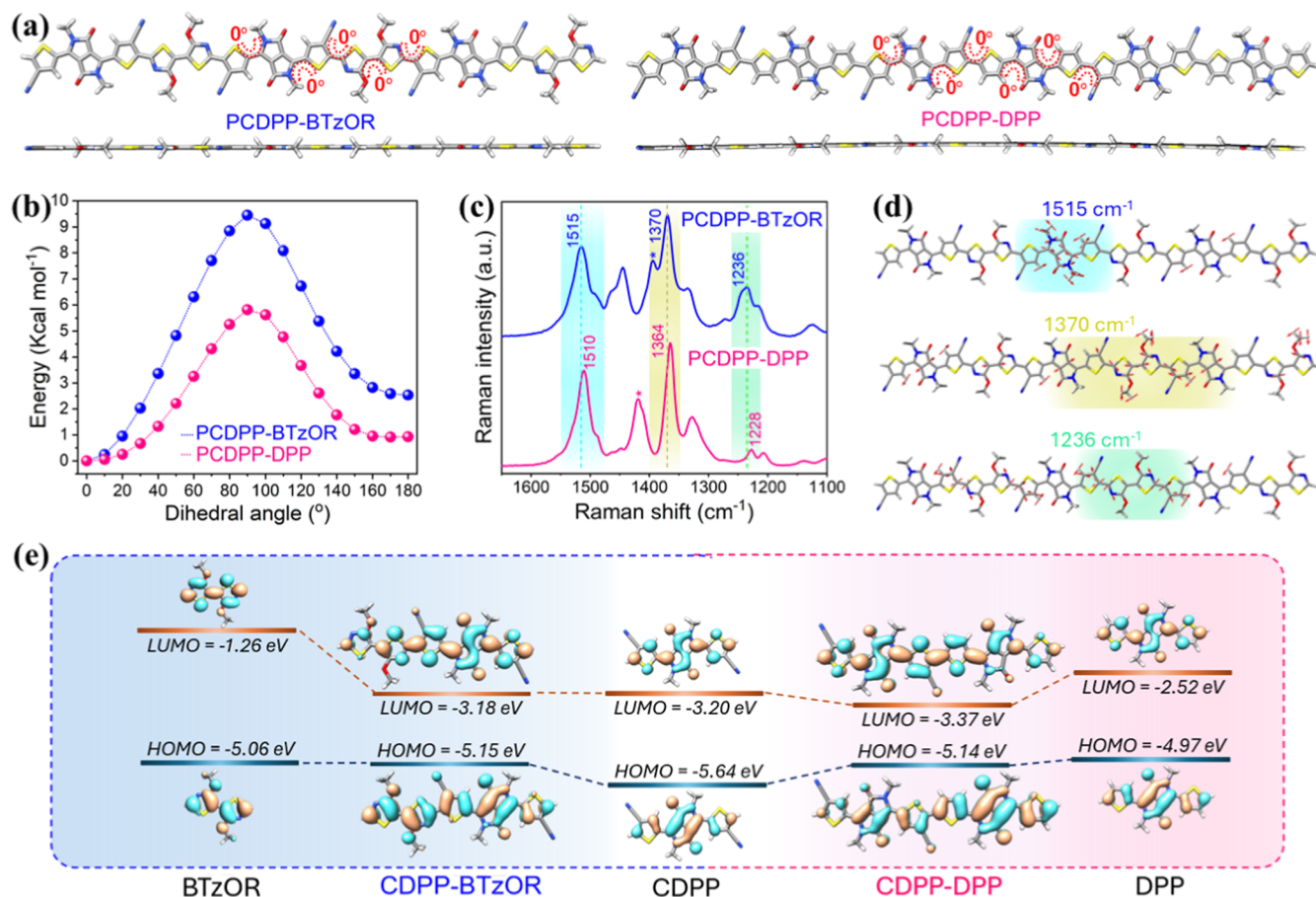


Figure 1. (a) Optimized molecular geometries of trimeric models of polymers **PCDPP-BTzOR** and **PCDPP-DPP** at the B3LYP-D3/6–31G** level. The inter-ring dihedral angles between nonfused moieties are also shown. In all calculations, the side chains were replaced with methyl groups to reduce the computational cost. (b) Potential energy surface for the dihedral angle connecting both CDPP and either BTzOR or DPP counits. (c) Raman spectra of polymer films under 532 nm excitation. Peaks marked with an asterisk (*) correspond to vibrational modes localized on the BTzOR and DPP counits of **PCDPP-BTzOR** and **PCDPP-DPP**, respectively. (d) Vibrational eigenvectors associated with the most outstanding C=C/C–C normal modes of **PCDPP-BTzOR** as representative example. (e) FMO energy diagram of the CDPP-based monomeric units. Calculations were performed at the B3LYP-D3/6–31G** level of theory. HOMO levels are shown as blue lines, and LUMO levels as red lines.

S cm⁻¹, $PF = 23.52 \mu W m^{-1} K^{-2}$) over **PCDPP-BTzOR** ($\sigma = 1.97 S$ cm⁻¹, $PF = 2.32 \mu W m^{-1} K^{-2}$). Owing to its high PF and excellent thermal insulation, **PCDPP-DPP** achieves a promising room-temperature thermoelectric figure-of-merit ($ZT = 0.07$). These results highlight the critical role of deliberate orbital interaction tuning in the development of high-performance n-type polymer thermoelectrics.

2. RESULTS AND DISCUSSION

2.1. Polymer Synthesis and Thermal Properties

Synthetic routes to the two polymers investigated in this study are illustrated in Scheme 1, with detailed experimental procedures provided in the Supporting Information. In brief, the electron-deficient dibrominated monomer CDPP-2Br was

synthesized following a succinate-based protocol previously reported by some of us.³⁴ CDPP-2Br was then copolymerized via Stille coupling with two different bistrimethyltin comonomers, a diketopyrrolopyrrole-based unit (DPP-2Sn) and a dialkoxy bithiazole-based unit (BTzOR-2Sn), to afford the novel polymers **PCDPP-DPP** and **PCDPP-BTzOR**. To ensure high purity and optimal performance in subsequent device fabrication, the crude polymers were subjected to Soxhlet extraction with methanol, acetone, hexane and dichloromethane to remove low-molecular-weight fractions and residual impurities. The final dichloromethane fractions were redissolved in chloroform and reprecipitated into methanol, yielding the target polymers for comprehensive characterization and device integration. The chemical structures of the

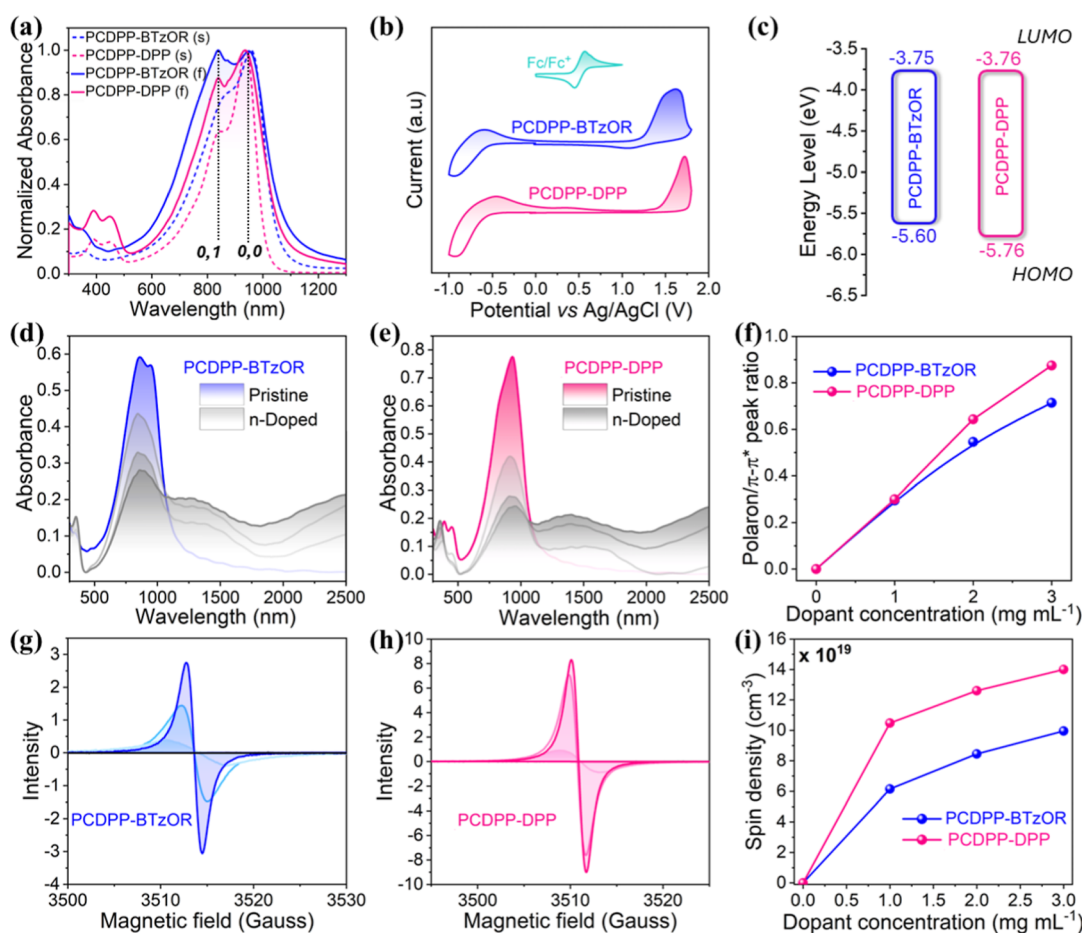


Figure 2. (a) Normalized UV-vis-NIR absorption spectra of **PCDPP-BTzOR** (blue lines) and **PCDPP-DPP** (pink lines) in both diluted chloroform solutions (dashed lines) and as thin films (solid lines). (b) CV curves of the two polymers, using the ferrocene/ferrocenium (Fc/Fc^+) redox couple as the external reference. (c) FMO level diagram derived from CV measurements. UV-vis-NIR absorption spectra of (d) **PCDPP-BTzOR** and (e) **PCDPP-DPP** films before and after n-doping with different concentrations of JLBI (1, 2, and 3 mg mL^{-1}). (f) Polaron/neutral peaks intensity ratio of polymer films n-doped with varying concentrations of JLBI. ESR signals of (g) **PCDPP-BTzOR** and (h) **PCDPP-DPP** films before and after n-doping. (i) Spin densities of polymer films n-doped with different concentrations of JLBI, extracted from the ESR signals shown above.

resulting polymers were confirmed by nuclear magnetic resonance (NMR) spectroscopy (Figures S1–S8). High-temperature gel permeation chromatography (HT-GPC) performed at 150 °C in trichlorobenzene revealed number-average molecular weights (M_n) of 39.3 kDa for **PCDPP-DPP** and 47.6 kDa for **PCDPP-BTzOR**, with dispersity index values (D) of 2.07 and 2.70, respectively (Table 1, Figures S9 and S10). These results indicate moderate molecular weights with acceptable dispersity, reflecting effective control over the polymerization process. Interestingly, thermogravimetric analysis (TGA) demonstrated an excellent thermal robustness for these materials, with decomposition temperatures at 5% weight loss exceeding 350 °C (Figure S11). Additionally, differential scanning calorimetry (DSC) revealed no apparent phase transitions for any of the polymers within the temperature range of 30–300 °C (Figure S12). Overall, these findings confirm the successful synthesis of two structurally related yet distinct conjugated polymers that combine high thermal stability, controlled molecular weight distribution, and excellent solution processability, key prerequisites for achieving reliable device performance and scalable fabrication.

2.2. Structural Characteristics

To gain insight into the influence of different monomer counts on the structural and electronic properties of the newly synthesized polymers, density functional theory (DFT)-based calculations were performed on trimeric models of these polymers at the B3LYP-D3/6–31G** level using the Gaussian 16 software. As shown in Figure 1a, theoretical modeling reveals that both **PCDPP-DPP** and **PCDPP-BTzOR** adopt fully coplanar polymer backbones, presumably facilitated by effective through-space intramolecular $\text{N}\cdots\text{S}$ interactions between the lone electron pairs of nitrogen atoms in the cyano groups of the CDPP moiety and the sulfur atoms of adjacent thiophene/thiazole units. In fact, the calculated $\text{N}\cdots\text{S}$ distance of 3.34 Å is shorter than the sum of their van der Waals radii (3.40 Å), indicating the presence of stabilizing noncovalent interactions that help conformationally lock the polymer backbone into a planar geometry. Remarkably, this backbone rigidity is further reinforced in **PCDPP-BTzOR** by additional $\text{N}\cdots\text{S}$ interactions between the nitrogen atoms of the BTzOR unit and the sulfur atoms of neighboring thiophenes within the CDPP moiety, with short $\text{N}\cdots\text{S}$ distances of 2.97 Å. This enhanced intramolecular locking is consistent with the higher torsional energy barrier calculated for rotation between

CDPP and BTzOR counits in **PCDPP-BTzOR** relative to that in the CDPP-DPP linkage of **PCDPP-DPP** (Figures 1b and S13–S14).

Beyond structural considerations, the conformational disparities in these systems are expected to influence electronic delocalization, an intriguing aspect requiring deeper investigation. To this end, Raman spectroscopy was employed to elucidate the relationship between the different counits and the degree of π -conjugation in these polymers. As shown in Figure 1c, the Raman spectra of **PCDPP-BTzOR** and **PCDPP-DPP** films under 532 nm excitation display pronounced differences, reflecting distinct extents of intramolecular π -electron delocalization.^{35,36} We first focus on the Raman band at approximately 1510 cm^{-1} (highlighted in cyan), which corresponds to a normal mode mostly located on the CDPP units (Figure 1d), as evidenced by the eigenvectors in Figures S16 and S17. Notably, this vibration exhibits a clear downshift of 5 cm^{-1} when transitioning from **PCDPP-BTzOR** to **PCDPP-DPP**. Since both polymers share the same CDPP unit, one could attribute such downshift to a more effective electronic delocalization within the π -system of **PCDPP-DPP**, which reduces the force constant of the associated bond-stretching vibration. Further features in the Raman spectra reinforce this interpretation: (i) The band at around 1370 cm^{-1} , assigned to the backbone-delocalized C=C/C–C stretching mode (highlighted in gold in Figure 1d), is downshifted by 6 cm^{-1} in **PCDPP-DPP** relative to **PCDPP-BTzOR**. (ii) The band near 1235 cm^{-1} (highlighted in green), associated with normal modes localized on the cyanated thiophene of the CDPP unit and adjacent thiophene or thiazole moieties, experiences an even larger downshift of 8 cm^{-1} in **PCDPP-DPP**. These collective downshifts suggest a more efficient intrachain electronic coupling between the CDPP and DPP counits, leading to an enhanced π -electron delocalization in **PCDPP-DPP** (see Supplementary Note S2).

This finding is further supported by the DFT-calculated FMO energy diagrams of the CDPP-based monomeric units (Figure 1e). Although the HOMO and LUMO in both **CDPP-BTzOR** and **CDPP-DPP** monomers are delocalized over the entire molecule, their fragment-resolved electronic character differs markedly. In **CDPP-BTzOR**, the HOMO energy (−5.15 eV) closely matches that of the BTzOR fragment (−5.06 eV), while its LUMO (−3.18 eV) aligns almost perfectly with that of the CDPP fragment (−3.20 eV). Such small offsets between the combined and the fragment orbitals suggest minimal energy loss upon orbital mixing, indicative of an efficient intramolecular charge-transfer pathway. By contrast, **CDPP-DPP** displays stronger orbital hybridization between fragments: the monomer LUMO (−3.37 eV) lies below both fragment LUMOs (−2.52 and −3.20 eV for DPP and CDPP, respectively), and the HOMO (−5.14 eV) is intermediate between those of the CDPP (−5.64 eV) and DPP (−4.97 eV) fragments. These energetic patterns reveal that **CDPP-BTzOR** has a greater donor–acceptor polarity and a more pronounced ICT character, whereas **CDPP-DPP** displays stronger orbital mixing and less polarized charge-transfer excitations (Figure S15).

2.3. Optoelectronic Properties

To obtain deeper insight into the electronic properties of these polymers, ultraviolet–visible–near-infrared (UV–vis–NIR) absorption spectra were recorded in both chloroform solution and thin-film states. As seen in Figure 2a, both polymers show

characteristic dual-band absorption profiles consisting of a weak high-energy band below 500 nm and a strong, shoulder-shaped low-energy band that extends broadly into the near-infrared region (700–1200 nm),^{37,38} typical of DPP-based systems. According to time-dependent (TD)-DFT calculations, the dominant low-energy transition ($S_0 \rightarrow S_1$) is attributed to a π – π^* excitation from the HOMO to the LUMO, which are delocalized across most of the polymer backbone as expected for highly π -conjugated architectures (Figure S19).³⁹ In diluted chloroform solution, incorporation of the BTzOR cunit in **PCDPP-BTzOR** induces a notable bathochromic shift of 18 nm in the absorption maximum (λ^{sol}) relative to **PCDPP-DPP**, from 938 to 956 nm. A similar effect is observed in the solid state, with **PCDPP-BTzOR** exhibiting a 12 nm red-shifted absorption maximum (λ^{film}) relative to **PCDPP-DPP**. This pronounced shift, accompanied by appreciable band broadening, further suggests ICT interaction between the CDPP moiety and the BTzOR unit, which can modulate the material's charge transport properties. Closer scrutiny reveals that both copolymers exhibit well-structured vibronic features in the low-energy absorption band, with two obvious 0–0 and 0–1 peaks. Notably, the relative intensity of the 0–1 peak was found to increase in **PCDPP-BTzOR**, indicative of enhanced interchain interactions presumably promoted by its enhanced ICT character that favors π – π stacking compared with **PCDPP-DPP**.^{40–42} In fact, temperature-dependent absorption spectra further reveal that although both polymers display partial aggregated features in solution (Figure S20), **PCDPP-BTzOR** undergoes more pronounced hypsochromic shifts and band narrowing upon heating.

These features suggest stronger molecular aggregation and more entangled polymer networks in **PCDPP-BTzOR**, as later confirmed by 2D grazing incident wide-angle X-ray scattering (2D-GIWAXS) analysis. This behavior is mirrored in the solid state, where both polymers exhibit significantly enhanced 0–1 peaks due to the formation of more ordered intermolecular arrangements in the film morphology. Curiously, the spectral shifts upon film formation are most pronounced for **PCDPP-BTzOR** (Table 1), further supporting its stronger π – π stacking propensity compared with **PCDPP-DPP**. Despite these structural differences, both polymer films exhibit comparable optical bandgaps ($E_{\text{g}}^{\text{opt}}$), with values of 1.12 and 1.14 eV for **PCDPP-BTzOR** and **PCDPP-DPP**, respectively.

To clarify the influence of the cunit on FMOs, the HOMO and LUMO energy levels were determined by cyclic voltammetry (CV), with the corresponding values summarized in Table 1. As shown in Figure 2b, both polymers demonstrate amphoteric redox behavior, displaying considerable oxidation and reduction peaks. However, incorporation of the DPP cunit attenuates the oxidation peak and shifts the oxidation potential to higher values while leaving the reduction potential unchanged compared to its BTzOR-based analogue, suggesting a suppressed oxidative character that likely translates into less pronounced ambipolar behavior. According to the reduction and oxidation onsets, the LUMO/HOMO energy levels of **PCDPP-BTzOR** and **PCDPP-DPP** were estimated to be −3.75/−5.60 eV and −3.76/−5.76 eV, respectively, corresponding to electrochemical bandgaps of 1.85 and 2.00 eV (Figure 2c). These values are in good agreement with the trend observed for the $E_{\text{g}}^{\text{opt}}$. Notably, **PCDPP-DPP** exhibits the deepest-lying HOMO, a feature that likely suppresses hole accumulation in OTFT devices.

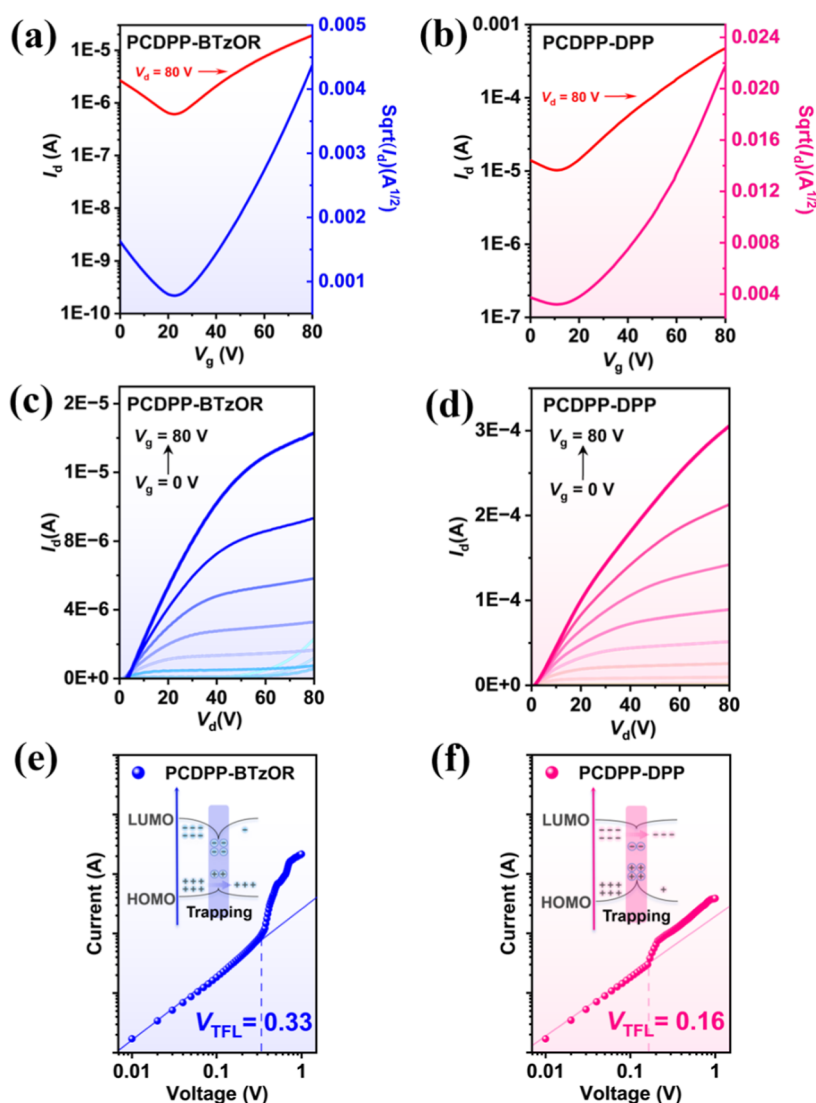


Figure 3. (a,b) Transfer and (c,d) output characteristics of OTFTs based on (a,c) PCDPP-BTzOR and (b,d) PCDPP-DPP. In the output plots, the gate voltage (V_g) was increased from 0 to 80 V in 10 V steps. All curves were obtained from devices with a channel length of 10 μm and channel width of 5 mm. (e,f) Trap density characterization of electron-only devices based on (e) PCDPP-BTzOR and (f) PCDPP-DPP using space-charge-limited current analysis.

2.4. n-Doping Characteristics

Given the critical role of doping efficiency in determining the performance of electronic devices, a preliminary spectroscopic-assisted evaluation of the n-dopability of these new polymers was carried out. To this end, both polymers were n-doped using a sequential doping protocol assisted by the transition-metal-catalyzed doping technique (Figure S21),⁴³ employing the julolidine-functionalized benzimidazole-based molecular dopant JLBI given its excellent solution processability, strong reducing capacity and minimal microstructure disruption during n-doping.^{44,45} In this context, UV-vis-NIR absorption was initially employed to monitor polaron formation as the JLBI concentration increased up to 3 mg mL^{-1} (Figure 2d,e). Even at a relatively low loading of 1 mg mL^{-1} , both polymers exhibited clear signatures of n-doping, as evidenced by the progressive bleaching of the neutral absorption bands in the 600–1200 nm region as well as the emergence of two prominent sub-bandgap polaronic features: a P1 band localized around 1500 nm (assigned to the HOMO-SOMO transition) and a low-energy P2 band localized in the near-infrared region

(assigned to the SOMO-LUMO transition). Since efficient doping is expected to yield stronger polaronic and weaker $\pi-\pi^*$ absorption at identical dopant loadings, the relative doping efficiencies of the polymers were assessed by tracking the polaron/ $\pi-\pi^*$ intensity ratio as a function of JLBI concentration (Figure 2f). In this sense, PCDPP-DPP displayed a slightly higher polaron/ $\pi-\pi^*$ ratio than PCDPP-BTzOR at comparable JLBI concentrations, qualitatively indicating more effective charge generation (Figure S22). This trend was reinforced by electron spin resonance (ESR) spectroscopy. As expected, pristine films showed no detectable ESR signal, while progressive increases were observed with higher dopant loadings as a consequence of the growing population of charge carriers (Figure 2g,h). Consistently, JLBI-doped PCDPP-DPP films exhibited stronger ESR signals than PCDPP-BTzOR, thereby confirming its superior doping efficiency (Figure 2i). Integration of the ESR signals further allows quantification of the density of unpaired spins; at a JLBI concentration of 3 mg mL^{-1} , spin densities of $\sim 1.4 \times 10^{20}$ and $\sim 9 \times 10^{19} \text{ cm}^{-3}$ were determined for PCDPP-DPP and

Table 2. OTFT and OTE Performance Parameters of the Newly Synthesized PCDPP-Based n-Type Polymers

polymer	OTFT			
	$\mu_{\text{e,sat}}$ [cm ² V ⁻¹ s ⁻¹] ^a	V_{Th} [V] ^b	$I_{\text{on}}/I_{\text{off}}$	
PCDPP-BTzOR	0.013 (0.009 ± 0.003)	31.5 ± 2.5	2 × 10 ¹	
PCDPP-DPP	0.191 (0.181 ± 0.011)	30.7 ± 1.4	5 × 10 ¹	
polymer	OTE			
	σ [S cm ⁻¹] ^c	S [μV K ⁻¹] ^d	PF [μW m ⁻¹ K ⁻²]	ZT
PCDPP-BTzOR	1.97 (1.87 ± 0.03)	−123.21 (−120.60 ± 2.74)	2.32 (2.21 ± 0.11)	-
PCDPP-DPP	25.71 (24.09 ± 0.73)	−111.02 (−109.10 ± 2.80)	23.52 (22.67 ± 0.43)	0.07 (0.06 ± 0.005)

^aMaximum electron mobilities. ^bAverage threshold voltage values. ^cMaximum electrical conductivities. ^dSeebeck coefficients corresponding to the highest PFs. The average values obtained from 5 devices together with the standard variation are included in parentheses.

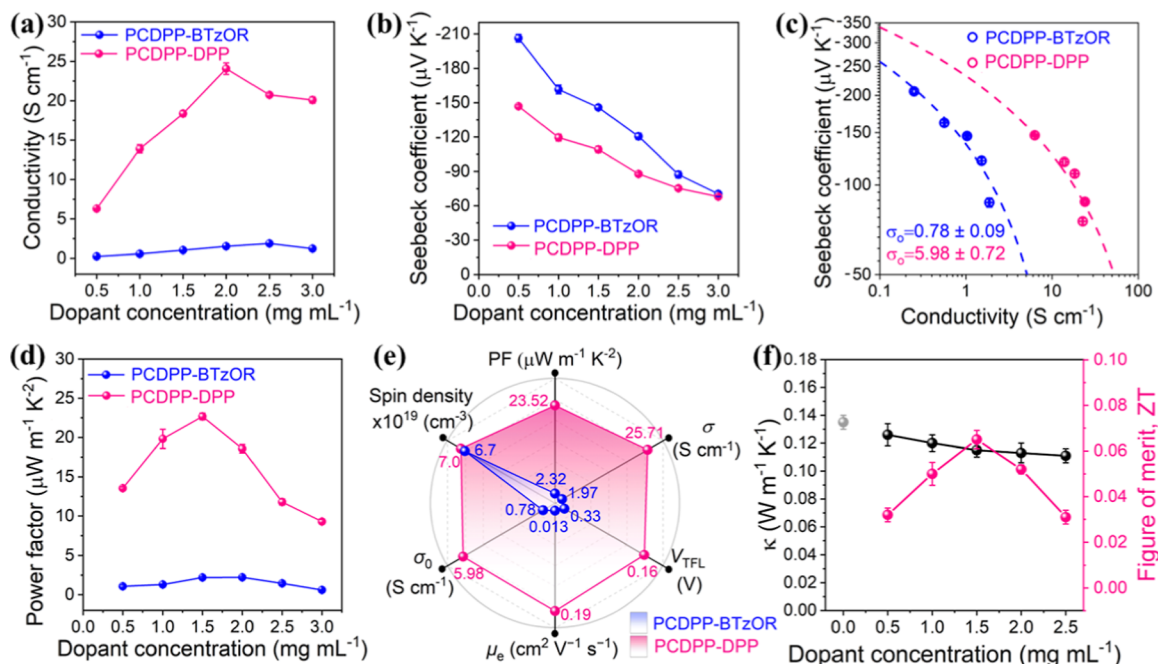


Figure 4. (a) Average electrical conductivities and (b) Seebeck coefficients of PCDPP-BTzOR (blue) and PCDPP-DPP (pink) films doped with different concentrations of JLBI. (c) S - σ relationship for the n-doped PCDPP-based films, fitted using the power-law model proposed by Kang and Snyder. The determined transport coefficients are also indicated. (d) Corresponding power factors calculated by the equation $PF = \sigma S^2$. (e) Radar chart comparing key electrical and thermoelectric parameters of PCDPP-BTzOR (blue) and PCDPP-DPP (pink), including the highest spin concentration, electrical conductivity and power factor of n-doped films; transport coefficients; as well as the highest electron mobilities and trap-filled limit voltages measured in OTFTs. (f) Thermal conductivity (black line) and thermoelectric figure of merit ZT (pink line) of n-doped PCDPP-DPP films as a function of JLBI concentration. The gray dot indicates the thermal conductivity of the pristine PCDPP-DPP film. Reported data corresponds to average values from at least five devices, with error bars representing the standard deviation.

PCDPP-BTzOR, respectively. These findings demonstrate that, despite their comparable LUMO energies, the two polymers show notably different n-doping efficiencies. This intriguing disparity highlights the importance of polymer-dopant miscibility, a key factor that will be examined in detail below.

2.5. Charge Transport Properties

Prompted by the excellent n-dopability demonstrated by the newly synthesized polymers, we next turned our attention to their intrinsic charge transport properties as a key parameter directly determining electrical conductivity ($\sigma = n\mu e$). To this end, OTFTs were fabricated using a top-gate/bottom-contact (TG/BC) device architecture of glass/Au/CsF/semiconductor/CYTOP/Al, with the amorphous fluoropolymer CYTOP serving as the gate dielectric layer (see Supporting Information for detailed fabrication and characterization procedures). Representative transfer and output characteristics of PCDPP-DPP and PCDPP-BTzOR are illustrated in Figure 3a,d, with

the corresponding device performance parameters summarized in Tables 2 and S1.

It is worth noting that most reported TDPP-based polymers exhibit ambipolar or even predominantly p-type transport, primarily due to the presence of two electron-donating thiophene flanking units within the TDPP framework and the direct conjugation of the nitrogen lone pairs from the amide groups with the polymer backbone.^{28,30} Indeed, the reference TDPP-based homopolymer, lacking the electron-withdrawing CN substituents incorporated in PCDPP-DPP, typically exhibits clear p-type transport with low mobility of 1×10^{-4} cm² V⁻¹ s⁻¹.^{46,47} In contrast, both PCDPP-DPP and PCDPP-BTzOR exhibited V-shaped transfer curves as well as superlinear current increase at low gate voltage (V_g) and high source-drain bias (V_d), indicative of minority hole injection and ambipolar transport (Figure 3a). Notably, PCDPP-DPP displays a reduced degree of ambipolarity, with predominantly n-type transport characterized by a low off-current ($\sim 10^{-6}$ A)

and an on/off current ratio ($I_{\text{on}}/I_{\text{off}} \approx 10^2$) (Figure 3b). This behavior can be attributed to more effective stabilization of both the LUMO and HOMO energy levels enabled by the DPP count. As a result, PCDPP-DPP-based OTFTs yielded a maximum μ_e of $0.19 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and a V_{Th} of 30 V. By comparison, the PCDPP-BTzOR analogue exhibits higher minority hole injection and more pronounced ambipolar transport (Figure 3a), which likely promote electron trapping. This interpretation is further supported by space-charge-limited current (SCLC) measurements on electron-only devices of these materials (Figure 3e,f). The trap-filled limit voltage (V_{TFL}) for PCDPP-BTzOR was found to be 0.33 V, double of that of PCDPP-DPP (0.16 V), revealing a significantly higher trap density. Consequently, PCDPP-BTzOR exhibited an order of magnitude lower μ_e of $0.013 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ together with a comparable V_{Th} of 32 V, a finding that clearly aligns with previous studies linking trap density to reduced carrier mobility.^{48,49}

As a first instance, the significantly reduced charge-transport performance of PCDPP-BTzOR appears somewhat counter-intuitive given its higher-lying LUMO energy level, more planar backbone, extended π -conjugation, and greater molecular weight compared to PCDPP-DPP. This apparent discrepancy emphasizes a crucial yet vaguely explored design principle, where achieving high-performance unipolar n-type polymers requires more than simply increasing electron deficiency, backbone planarity or conjugation length. Instead, it critically relies on the rational selection of comonomer units with well-matched electronic properties that enable efficient charge-transport pathways along the polymer backbone while suppressing charge trapping effects. In this context, the incorporation of the BTzOR count in PCDPP-BTzOR results in less efficient transport compared to PCDPP-DPP due to its enhanced donor–acceptor character. Specifically, the BTzOR count raises the HOMO energy of PCDPP-BTzOR (−5.60 eV) relative to PCDPP-DPP (−5.76 eV), making it less stabilized against hole injection (Figure S25). The increased susceptibility to hole injection promotes recombination and trap-assisted transport, which ultimately limits effective n-type charge transport and suppresses electron mobility.

2.6. Thermoelectric Performance

Building on our insights into n-doping efficiency and intrinsic charge transport properties, we systematically investigated how these parameters govern the overall thermoelectric performance of the PCDPP-based polymers under controlled doping conditions. To mitigate $\text{O}_2/\text{H}_2\text{O}$ -induced dedoping and ensure reproducibility, thermoelectric devices were fabricated and characterized in an N_2 -filled glovebox using a sequential doping protocol catalyzed by transition metals (see Supporting Information for details).⁴³ The key performance parameters are summarized in Figure 4 and Table 2. As shown in Figure 4a, both polymers exhibit very low intrinsic conductivity. Upon increasing the JLBI dopant concentration, σ was substantially enhanced due to the generation of mobile charge carriers, reaching peak values of $24.09 \pm 0.73 \text{ S cm}^{-1}$ for PCDPP-DPP at 2 mg mL^{-1} and $1.87 \pm 0.03 \text{ S cm}^{-1}$ for PCDPP-BTzOR at 2.5 mg mL^{-1} . Notably, the σ of JLBI-doped PCDPP-BTzOR remains over 1 order of magnitude lower than that of PCDPP-DPP. Given that $\sigma = n\mu_e$, this trend is consistent with the lower μ_e measured in OTFTs for PCDPP-BTzOR (0.013 vs $0.19 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for PCDPP-DPP), as well as its approximately half carrier concentration at similar dopant loadings. Interestingly,

further increasing the JLBI concentration to 3 mg mL^{-1} led to a gradual decline in σ to $20.10 \pm 0.41 \text{ S cm}^{-1}$ for PCDPP-DPP and $1.22 \pm 0.02 \text{ S cm}^{-1}$ for PCDPP-BTzOR, presumably due to the interplay of carrier–carrier repulsion and dopant-induced microstructural disorder in heavily doped films.^{50,51} Importantly, similar performance trends were observed when using the widely employed N-DMBI molecular dopant, although the resulting conductivities were slightly lower than those obtained with JLBI (see Figure S24). To evaluate thermoelectric performance, Seebeck coefficients were measured by recording the thermovoltage generated across the n-doped films under applied temperature gradients from 0 to 2.8 K in 0.7 K increments (Figures S26 and S28). As expected, all doped films exhibited negative S values, confirming n-type character dominated by electron transport. Furthermore, the absolute values of S ($|S|$) decreased progressively with increasing dopant concentration from 0.5 to 3 mg mL^{-1} , reflecting the typical inverse correlation between $|S|$ and carrier concentration (Figures 4b and S27).^{14,52} For PCDPP-DPP, $|S|$ decreased from -146.77 ± 1.26 to $-68.04 \pm 1.30 \mu\text{V K}^{-1}$, while for PCDPP-BTzOR it decreased from -206.29 ± 3.37 to $-68.59 \pm 1.60 \mu\text{V K}^{-1}$. However, a deeper analysis of the S - σ relationship revealed a less pronounced drop in $|S|$ with increasing σ for PCDPP-DPP, thereby shifting its S - σ curve upward relative to that of PCDPP-BTzOR (Figure 4c). For instance, at a dopant concentration of 3 mg mL^{-1} , despite similar S values of approximately $-70 \mu\text{V K}^{-1}$ were observed for both polymers, PCDPP-DPP yielded a markedly higher σ than PCDPP-BTzOR (20.09 ± 0.41 vs $1.22 \pm 0.02 \text{ S cm}^{-1}$). Fitting the S - σ data to the power-law model proposed by Kang and Snyder, with a transport parameter of $s = 1$ as observed in comparable systems,^{53,54} reveals a higher transport coefficient ($\sigma_0 = 5.98 \pm 0.72$) for PCDPP-DPP compared to PCDPP-BTzOR ($\sigma_0 = 0.78 \pm 0.09$).^{54,55} These findings indicate that the enhanced σ observed in n-doped PCDPP-DPP films is mostly driven by well-preserved and superior charge carrier mobility in the presence of the JLBI dopant.⁵⁴ As a consequence of simultaneously enhancing both σ and S , PCDPP-DPP reached an excellent PF of $22.67 \pm 0.43 \mu\text{W m}^{-1} \text{ K}^{-2}$ at 1.5 mg mL^{-1} of JLBI, which is 10-fold higher than the PF of $2.21 \pm 0.11 \mu\text{W m}^{-1} \text{ K}^{-2}$ measured for PCDPP-BTzOR (Figure 4d and Table S2).

Inspired by the striking PF observed in n-doped PCDPP-DPP films, we proceeded to measure the thermal conductivity (κ) at room temperature using the transient hot-wire method,^{56,57} allowing for a comprehensive evaluation of the thermoelectric figure of merit ($ZT = \sigma S^2 T / \kappa$). In this context, the pristine PCDPP-DPP film displayed a low κ of $0.135 \pm 0.005 \text{ W m}^{-1} \text{ K}^{-1}$ at 300 K (Figure 4f), highlighting its excellent thermal insulation compared to typical n-type organic semiconductors, which typically exhibit values exceeding $0.2 \text{ W m}^{-1} \text{ K}^{-1}$.^{6,9,58} Interestingly, κ decreased further with increasing the JLBI concentration due to enhanced phonon scattering by free carriers and dopant-induced disruptions in molecular packing,^{52,59,60} reaching a minimum of $0.111 \pm 0.006 \text{ W m}^{-1} \text{ K}^{-1}$ at the highest dopant loading (2.5 mg mL^{-1}). Notably, the most balanced combination of PF ($22.67 \pm 0.43 \mu\text{W m}^{-1} \text{ K}^{-2}$) and κ ($0.115 \pm 0.007 \text{ W m}^{-1} \text{ K}^{-1}$) was obtained at a JLBI dopant concentration of 1.5 mg mL^{-1} , resulting in a high ZT value of 0.06 ± 0.0005 at room temperature. To our knowledge, this value is among the highest room-temperature ZT reported for purely n-type organic thermo-

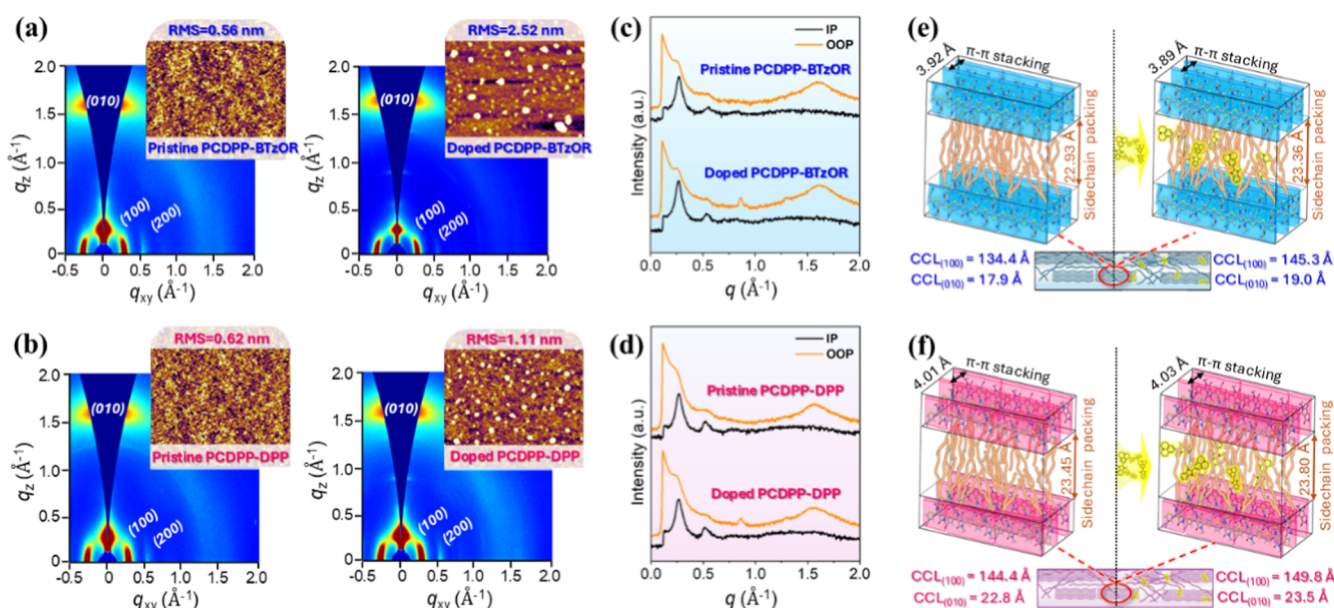


Figure 5. AFM height images and 2D-GIWAXS patterns of (a) PCDPP-BTzOR and (b) PCDPP-DPP polymer films in their pristine state and after n-doping. Corresponding in-plane (IP) and out-of-plane (OOP) line-cut profiles are shown in (c) for PCDPP-BTzOR and (d) for PCDPP-DPP. Schematic representations of the microstructural changes induced in (e) PCDPP-BTzOR and (f) PCDPP-DPP upon n-doping with JLBI (highlighted in yellow).

electronics,^{8,61–66} firmly establishing PCDPP-DPP as a promising design motif for the next-generation plastic electronics.

2.7. Thin-Film Microstructure and Morphology

The above results triggered our interest in clarifying the impact of the structural modification on polymer film morphology and microstructure, which was initially investigated through atomic force microscopy (AFM) characterizations of the films prepared under the same conditions used for optimal device fabrications. As shown in Figures S31 and S32, both PCDPP-BTzOR and PCDPP-DPP films exhibit finely interpenetrating networks with exceptionally smooth surfaces, as reflected in their low root-mean-square (RMS) roughness values of ~ 0.5 nm. Specifically, RMS values of 0.56 nm for PCDPP-BTzOR (Figure 5a) and 0.62 nm for PCDPP-DPP (Figure 5b) indicate minimal surface irregularities and strongly suppressed grain boundaries after optimal annealing. However, a completely different scenario emerges after n-doping with JLBI, where obvious bright spots appear in the AFM height images presumably arising from dopant aggregation at the film surface due to the sequential doping strategy adopted.⁶⁷ This results in rougher surfaces, with RMS roughness increasing to 2.52 nm for PCDPP-BTzOR and 1.11 nm for PCDPP-DPP. Notably, the dopant-rich domains are found to be significantly smaller in doped PCDPP-DPP films, pointing to better miscibility between JLBI and this polymer.^{59,68} This improved miscibility is crucial for preserving film integrity and ensuring a more uniform dopant distribution within the polymer matrix,^{44,69} thereby contributing to the higher doping efficiency and superior thermoelectric performance of PCDPP-DPP.

To gain deeper insights into the supramolecular packing and molecular orientation of the PCDPP-based polymers, we carried out 2D-GIWAXS measurements on the corresponding thin films. Figure 5a,b show the 2D-GIWAXS patterns of PCDPP-BTzOR and PCDPP-DPP respectively, with the associated line-cut profiles presented in Figure 5c.

Scrutiny of the data reveals that both polymers predominantly adopt a face-on orientation, as evidenced by pronounced (100) lamellar diffraction peaks along the in-plane (IP) direction at $q_{xy} \approx 0.27 \text{ \AA}^{-1}$ and a distinctive (010) peak in the out-of-plane (OOP) direction at $q_z \approx 1.60 \text{ \AA}^{-1}$ arising from cofacial π - π stacking. In particular, the (100)/(010) diffraction peaks of pristine PCDPP-BTzOR and PCDPP-DPP films appear at q values of $0.274/1.605 \text{ \AA}^{-1}$ and $0.267/1.566 \text{ \AA}^{-1}$, corresponding to lamellar/ π - π stacking distances of 22.93/3.92 Å and 23.53/4.01 Å, respectively. These results indicate that PCDPP-BTzOR exhibits a more densely packed microstructure with reduced intermolecular spacing compared to PCDPP-DPP, presumably due to the less conformationally locked conjugated backbone of the latter. To quantitatively evaluate crystallinity differences, the crystal coherence lengths (CCLs) of the (100) and (010) reflections were estimated according to the Scherrer equation, which quantifies the distance over which crystalline order is preserved along specific directions.⁷⁰ As summarized in Table S3, the pristine films of PCDPP-BTzOR (134.35/17.92 Å) and PCDPP-DPP (144.44/22.75 Å) show comparable $CCL_{xy,(100)}/CCL_{z,(010)}$ values, thereby pointing to similar grain-boundary scattering and energy disorder during charge transport in both polymers. Taken together, these findings show that PCDPP-BTzOR features a denser supramolecular packing while maintaining crystalline domains comparable to those of PCDPP-DPP. Therefore, it seems reasonable to assume that its substantially lower μ_e observed in OTFTs (0.013 vs $0.19 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) cannot be attributed to microstructural limitations, but rather underscores the critical role of suppressing charge-trapping effects in determining device performance.

Since introducing molecular dopants into conjugated polymer matrices often causes significant structural disruption, we also investigated how n-doping with JLBI influences the microstructure of our materials. Interestingly, the diffraction patterns of both polymers remained nearly unchanged after n-

doping (bottom profiles of Figure 5c,d), suggesting that their microstructural order is well-preserved even at high dopant loadings (Figure S18). In fact, dopant permeation even slightly enhances film crystallinity along the π - π stacking direction by ≈ 1 Å, reaching $\text{CCL}_{z,(010)}$ values of 19.04 Å for PCDPP-BTzOR and 23.51 Å for PCDPP-DPP after n-doping. Curiously, while the π - π stacking distances were largely unaffected during the doping process (with only minor variations of ≈ 0.03 Å), the lamellar spacing increased up to ≈ 0.5 Å in JLBI-doped films of both polymers. We tentatively attribute this behavior to partial intercalation of dopant molecules between the alkyl side chains of the polymers, as schematically illustrated in Figure 5e,f. Consequently, the affinity of JLBI for the alkyl chains locally affects lamellar ordering through dopant-induced intermolecular interactions,^{61,71} as revealed by the $\text{CCL}_{xy,(100)}$ increase of 11.0 Å (5.5 Å) for PCDPP-BTzOR (PCDPP-DPP) compared to their pristine counterparts. The minimal structural impact observed in PCDPP-DPP, attributable to its enhanced miscibility with the JLBI dopant, combined with its efficient charge-transport pathways, accounts for its superior electrical conductivity and improved thermoelectric performance.

3. CONCLUSION

In summary, two new n-type conjugated polymers, PCDPP-DPP and PCDPP-BTzOR, were rationally designed and synthesized to investigate the scarcely explored yet critical role of orbital hybridization between different counits in determining thermoelectric performance. Despite PCDPP-BTzOR exhibiting a more conformationally locked backbone, higher molecular weight, comparable film crystallinity, and nearly identical LUMO level to its PCDPP-DPP analogue, the latter demonstrated markedly superior charge transport and doping response. Specifically, PCDPP-DPP achieved a higher μ_e in OTFTs (0.191 vs 0.013 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$) and a greater charge carrier concentration after n-doping (1.4×10^{20} vs 1.0×10^{20} spins cm^{-3}), resulting in over an order of magnitude higher σ in OTEs (25.71 vs 1.97 S cm^{-1}). These improvements yielded a high PF of 23.52 $\mu\text{W m}^{-1} \text{K}^{-2}$ for PCDPP-DPP, which combined with its excellent thermal insulation ($\kappa = 0.115 \text{ W m}^{-1} \text{K}^{-1}$), culminated in a promising room-temperature figure of merit of $\text{ZT} = 0.07$. To the best of our knowledge, this ZT value ranks among the highest reported for purely n-type polymers at room temperature, revealing that effective orbital hybridization between counits can be a more decisive factor governing charge transport than the conventionally pursued backbone planarity or LUMO-level engineering. This work thus demonstrates that deliberate tuning of interunit orbital interactions to enhance electronic delocalization while mitigating trapping associated with polarized charge-transfer states represents an effective strategy for designing high-performance n-type thermoelectric polymers.

4. EXPERIMENTAL SECTIONS

4.1. Materials

All chemicals and solvents were procured from Tokyo Chemical Industry (TCI) Co., Ltd., and used as received without further purification unless otherwise specified. The JLBI dopant was synthesized following previously reported procedures.^{44,45} The detailed synthetic procedures for both PCDPP-DPP and PCDPP-BTzOR are provided in the Supporting Information, together with complete NMR characterization data (^1H and ^{13}C spectra) for all monomers and the corresponding polymers.

4.2. Nuclear Magnetic Resonance (NMR) Spectra

NMR spectra were recorded on a Bruker Ascend 400 MHz spectrometer (^1H at 400 MHz, ^{13}C at 100 MHz) using CDCl_3 as the solvent.

4.3. Polymer Molecular Weights

Molecular weights were determined by high-temperature size-exclusion chromatography using a Polymer Laboratories PL-GPC 220 system (Agilent Technologies) at 150 $^\circ\text{C}$, with trichlorobenzene as eluent and polystyrene as standard for calibration.

4.4. Differential Scanning Calorimetry (DSC) and Thermogravimetric Analysis (TGA)

DSC and TGA were performed using a Mettler-Toledo STARe system under a nitrogen atmosphere at a heating rate of 10 $^\circ\text{C min}^{-1}$.

4.5. Cyclic Voltammetry (CV)

CV measurements were conducted on polymer films using a CHI660 potentiostat/galvanostat electrochemical workstation in a three-electrode configuration cell consisting of a Pt disk working electrode, a Pt wire counter electrode, and an Ag/AgCl reference electrode. Measurements were performed in 0.1 M tetra(*n*-butyl)ammonium hexafluorophosphate at a scan rate of 50 mV s^{-1} using the Fc/Fc⁺ couple as the external standard.

4.6. Theoretical Modeling

The optimum structures for the trimeric models of the polymers under study were optimized at the framework of the density functional theory (DFT) using the hybrid, generalized gradient approximation (GGA) functional B3LYP^{72,73} together with the 6-31G**^{74,75} basis set, as implemented in the GAUSSIAN16 program.⁷⁶ Note that these calculations include an explicit D3 correction for dispersion.⁷⁷ In all the cases, the large alkyl chains were substituted by methyl groups in order to reduce the computational cost.

A preliminary study of the dihedral potential energy hypersurface between the DPP unit and the adjacent thiophene or thiazole rings was carried out to better understand the conformation of these polymers. The relaxed potential energy hypersurfaces were computed by constraining the angle of the selected dihedral bonds (by step of 10 $^\circ$) and allowing all other degrees of freedom to relax to their potentially energy minimum. The resulting potential energy curve was then referenced to the minimal energy structure. Harmonic frequencies calculations were performed on the previously optimized molecular geometries and no imaginary frequencies were observed, ensuring the finding of the global minimum energy. Raman intensities were calculated by using an adjustment of the theoretical force fields in which the frequencies are scaled down by a factor of 0.965 to disentangle experimental misassignments. The theoretical spectra were obtained by convolution the scaled frequencies and the Raman scattering activities with Gaussian functions (10 cm^{-1} width at the half-height). Time-dependent DFT (TD-DFT) approach was used to assess the excited state vertical transition energies at the same level of theory.^{78,79} Molecular orbitals distribution and vibrational eigenvectors were plotted using the ChemCraft 1.8 molecular modeling software.

4.7. UV-Vis Absorption Spectra

We recorded the UV-vis absorption spectra of the polymers both in dilute chloroform solutions (10^{-4} M) and as thin films spin-coated onto glass substrates, using a Shimadzu UV-3600 spectrophotometer.

4.8. Raman Spectra

Raman spectra of the thin-film samples were acquired using a HORIBA LabRAM HR Evolution confocal Raman spectrometer with 532 nm laser excitation. The instrument is equipped with an 800 mm focal length monochromator, a confocal microscope featuring a $\times 100$ objective, and a high-performance CCD detector, providing a spectral resolution better than 1 cm^{-1} .

4.9. Electron Paramagnetic Resonance (EPR) Spectra

EPR spectra were recorded at X-band (9.84 GHz) on a Bruker EMX-plus-10 spectrometer under 2 G field modulation, 2 mW microwave power and 100 kHz modulation frequency. Polymer films were spin-coated onto PET substrates, cut into 2.5 mm × 10 mm strips inside a nitrogen-filled glovebox and sealed in quartz EPR tubes. Spin densities were quantified using Bruker software following standard procedures.

4.10. Atomic Force Microscopy (AFM)

AFM images of polymer films were recorded using a dimension icon scanning probe microscope (Asylum Research, MFP-3D-Stand Alone) working in tapping mode.

4.11. Two-Dimensional Grazing-Incidence Wide-Angle X-ray Scattering (2D-GIWAXS)

The thin films were deposited on silicon substrates and subsequently measured at the PLS-II 9A SAXS beamline of the Pohang Accelerator Laboratory, Republic of Korea.

4.12. Space-Charge-Limited Current (SCLC)

The curves were measured using electron-only devices with a configuration ITO/ZnO/Polymer/F₃N/Ag. The trap-filled limit voltage (V_{TFL}) was defined as the onset voltage of the trap-filled limit region.

4.13. Organic Thin-Film Transistors (OTFTs)

Top-gate/bottom-contact (TG/BC) devices were fabricated on 1.2 cm × 1.2 cm borosilicate glass substrates. Source and drain electrodes consisting of 30 nm Au/3 nm Cr with channel lengths (L) of 10, 20, 50, or 100 μ m and a channel width (W) of 5000 μ m were patterned by photolithography. The patterned glass substrates were cleaned by sequential sonication in isopropanol, acetone and isopropanol (20 min each), dried with a stream of N₂, and treated with UV–ozone for 15 min. After that, the substrates were transferred into the N₂-filled glovebox (O₂, H₂O concentration <0.1 ppm), where an electron injection layer of cesium fluoride (CsF, 2 mg mL^{−1} in 2-ethoxymethanol) was spin-coated at 1500 rpm for 30 s and annealed at 120 °C for 20 min. Active layers of PCDPP-BTzOR and PCDPP-DPP were spin-coated from chloroform solutions (5 mg mL^{−1}) at 1500 rpm/45 s and 800 rpm/45 s, respectively, followed by thermal annealing at 100 and 150 °C for 10 min. Subsequently, a ~500 nm thick CYTOP dielectric layer (CTL-809M:CT-SOLV180, 2:1 v/v, Asahi Glass Co. Ltd.) was spin-coated on top at 1500 rpm/60 s and annealed at 90 °C for 30 min. Finally, ~50 nm Al was thermally evaporated as the gate electrode to complete the device fabrication. OTFTs were characterized inside a N₂-filled glovebox using a Keithley 4200-SCS semiconductor analyzer. Saturation-regime mobilities (μ) were extracted from the standard equation

$$I_{SD} = (W/2L)\mu C_i(V_{SG} - V_{Th})^2$$

where I_{SD} is the source-drain current, C_i the capacitance per unit area of the insulating layer, V_{SG} the gate voltage, and V_{Th} the threshold voltage.

4.14. Organic Thermoelectric Devices (OTEs)

Borosilicate glass substrates were first sonicated in isopropanol, acetone and isopropanol again (30 min each), dried with a stream of N₂, and treated with UV–ozone for 15 min. After that, the substrates were transferred into a N₂-filled glovebox (O₂, H₂O concentration <0.1 ppm), where Au electrodes (30 nm thick, 100 μ m length and 2000 μ m width) were thermally evaporated through a shadow mask. To improve doping efficiency, ~0.1 nm of gold nanoparticles (Au-NPs) were thermally evaporated onto the substrates.⁴³ Subsequently, polymer solutions (5 mg mL^{−1} in chloroform) were spin-coated at 1000 rpm/45 s and annealed at 100 °C (PCDPP-BTzOR) or 150 °C (PCDPP-DPP) for 20 min, yielding average film thicknesses of 36 and 42 nm, respectively (Film thicknesses were measured using a Tencor KLA D-120 profilometer). For n -doping, a sequential doping protocol was used, in which JLBI dopant solutions (0.5–3 mg mL^{−1} in n -butyl acetate) were spin-coated over the semiconducting layer at 1500 rpm for 30 s, followed by thermal activation at 120 °C for 15 s.

Electrical conductivity was measured inside a N₂-filled glovebox with a Keithley 4200-SCS system by recording linear I – V curves, with average values calculated from five independent measurements. On the other hand, thermoelectric performance was measured using a pair of Peltier devices with a separation distance of 1 mm, in which temperature differences (ΔT) ranging from 0 to 2.8 K (in 0.7 K steps) were applied. The resulting thermal voltage was measured using a Keithley 4200-SCS system, and the Seebeck coefficients were determined from the slope of the linear fit of thermal voltage vs ΔT . Average Seebeck coefficients were calculated from five different measurements. Finally, thermal conductivity (κ) was measured using a XIATECH TC3000E instrument, which operates based on the transient hot wire (THW) method, an established technique known for its high accuracy and precision in measuring the thermal conductivity of thin films.⁸⁰ The instrument is equipped with a single metal wire sensor and requires only 60 s of measurement time to achieve highly reliable results.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.5c24154>.

Materials synthesis, computational details, doping and OTE characterization, morphologic characterization data and Raman spectra are included in the (PDF)

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Notes

The authors declare no competing financial interest.

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